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Team 10

ASL Alphabet Translation



Problem Statement

ASL Accessibility

62% of Deaf and **66%** of Hard-of-Hearing individuals reported that communication barriers impacted their career decisions, limiting job opportunities and professional growth. (Nagish, 2025)

Our Goal:

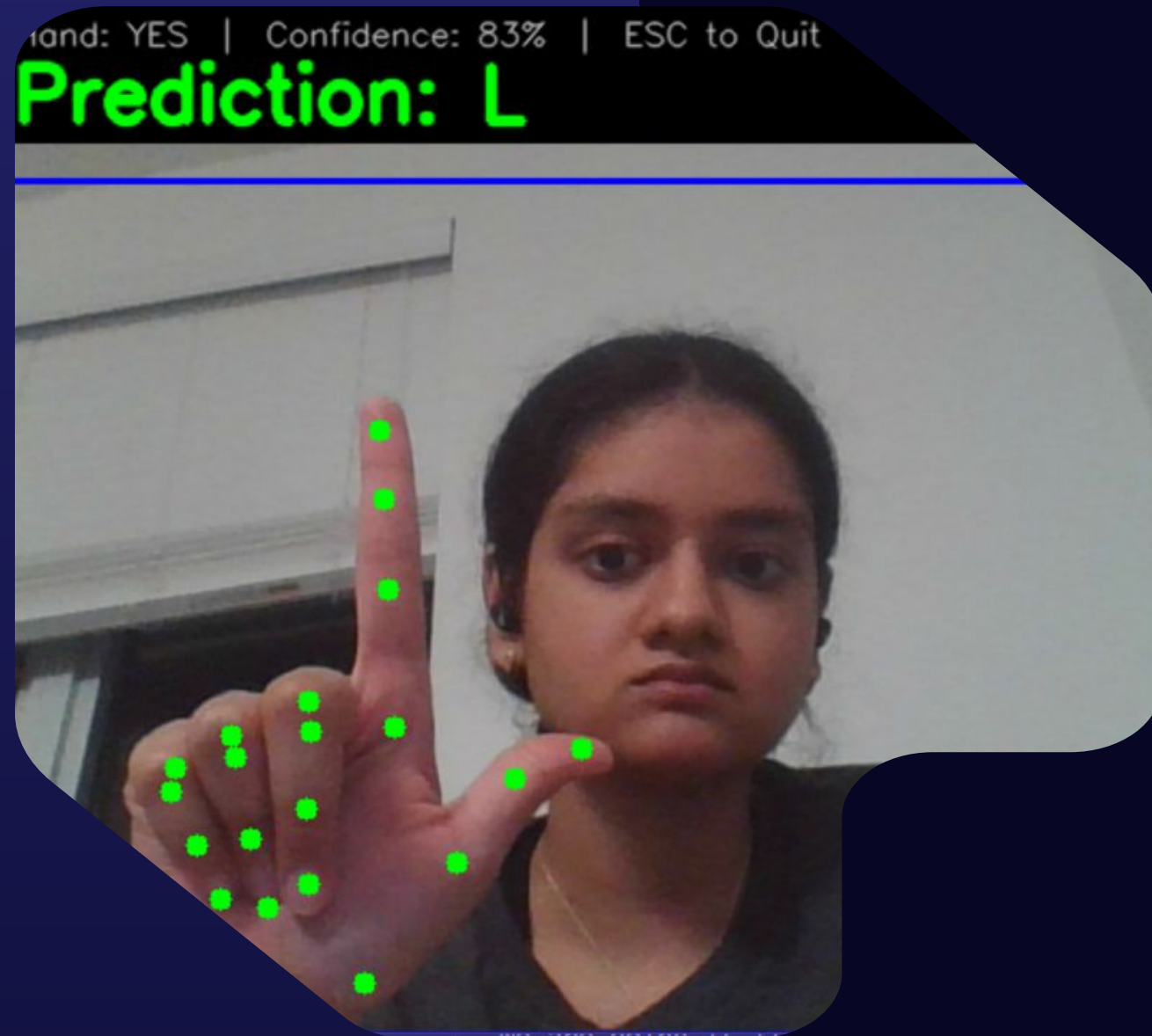
Build a real-time ASL alphabet translation system using a multi-stage computer vision pipeline that processes video input, to improve accessibility and enable seamless communication between ASL signers and non-signers.

Our Pipeline



Approach

- 1. YOLO** – detects and isolates the person, removing background noise for a clean region of interest
- 2. MediaPipe Hands** – extracts a 21-point skeletal representation, focusing only on hand structure
- 3. LSTM Network** – learns motion over time and classifies gestures into one of 26 ASL letters



Why This Works:

YOLO improves accuracy by isolating the person. Skeleton-based input keeps processing lightweight and efficient. LSTM captures motion, enabling action-based letters like J and Z. Diverse training data improves real-world robustness across users and environments.



Metrics



Evaluation

Target Metrics:

- Validation Accuracy: $\geq 95\%$
- Validation Loss: < 0.15
- Train Loss: < 0.10

Achieved Metrics:

- Validation Accuracy: 97.8%
- Validation Loss: 0.0775
- Train Loss: 0.0997

These metrics were obtained by training a 26-class sequence classifier on 4,676 samples over 50 epochs with learning rate scheduling

Work Cited



References

- Nagish. (2025). The Impact of Communication Technology in Empowering the Deaf and Hard-of-Hearing.
<https://www.prnewswire.com/news-releases/new-survey-reveals-ongoing-communication-barriers-for-deaf-and-hard-of-hearing-in-social-medical-and-professional-settings-302619991.html>



Live DEMO

Showcase

